

Empirical Design Decisions in Explainable Automotive Recommendation Systems: A Subspace Clustering Analysis

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Abstract

Automotive recommender systems face the unique challenge of matching users with highly complex items across fundamentally disparate semantic domains, including driving performance, technological features, safety ratings, and ownership costs. Traditionally, clustering pipelines concatenate these distinct domains into a single global feature space. In this paper, we investigate whether this monolithic representation obscures meaningful item structure through a phenomenon known as subspace masking. We benchmark global clustering against perspective-specific clustering on a real-world dataset of 251 vehicles described by 114 features. Utilizing KMeans, Gaussian Mixture Models, Agglomerative clustering, and HDBSCAN, our controlled experiments reveal that global clustering is primarily influenced by the largest feature block (Tech & Comfort), effectively masking the topological structure present in lower-dimensional domains like Safety and Ownership. Perspective-specific clustering consistently produces more coherent and stable groupings ($ARI > 0.90$). Furthermore, mechanism-testing ablations—including PCA reduction, random feature sampling, and variance-equalized representations—provide strong evidence that this degradation arises from cross-domain semantic interference rather than mere dimensionality imbalance. These findings suggest that semantic feature partitioning is a highly effective representation strategy for automotive datasets, establishing a coherent architectural foundation that motivates the future evaluation of downstream natural-language explanations.

1 Introduction

Automotive recommender systems face the unique challenge of matching users with highly complex, multi-dimensional items. A vehicle’s profile encompasses disparate domains, including driving performance, technological features, safety ratings, and ownership costs. Traditionally, clustering and recommendation pipelines concatenate these distinct domains into a single global feature space to model item similarity. However, combining low-dimensional domains (e.g., 7 ownership cost features) with high-dimensional domains (e.g., 81 technology features) introduces the risk of subspace masking, where distance metrics and probabilistic densities are disproportionately influenced by the most feature-dense categories.

In explainable AI (XAI) for recommender systems, capturing the specific intent of the user is critical. If a clustering algorithm fails to group vehicles by safety metrics due to the high variance of infotainment and convenience features, the underlying structure cannot accurately support recommendations based on a user’s safety preferences.

This paper makes four contributions:

1. A controlled comparison of clustering scope (global vs. perspective-scoped) fully crossed with clustering algorithm (KMeans, GMM, Agglomerative, HDBSCAN) on a real-world, mixed-domain automotive dataset.
2. A dimensionality-vs-semantics ablation (random feature sampling, PCA-reduced global baseline, and a variance-equalized ‘weighted global’ representation) that directly tests the feature-imbalance / subspace-masking mechanism hypothesis, rather than only observing a scope effect.
3. A stability and agreement analysis (bootstrap ARI/AMI, cross-perspective NMI, cluster balance entropy) characterizing how robust and how independent the discovered cluster structures are.
4. A methodologically corrected statistical testing protocol (Holm-Bonferroni adjusted pairwise comparisons, explicit small-sample Friedman caveats, bootstrap 95% CIs) for comparing clustering algorithms across conditions.

2 Related Work

2.1 Feature Representation and Multi-View Clustering

Traditional content-based recommender systems construct item profiles by concatenating disparate feature categories into a single, unified global feature space. While this simplifies the architectural pipeline, high-dimensional datasets often suffer from subspace masking, where subsets containing a disproportionately large number of features dominate the variance. To combat this, recent literature has shifted toward multi-view clustering, where different feature subsets

(“views”) of the same entities are clustered separately rather than jointly. Exploring joint-versus-separate feature-subset clustering allows models to capture localized semantic structures [7].

2.2 Cross-Domain Recommendation

In recommender systems, translating user preferences from one feature domain (e.g., Ownership) to another (e.g., Safety) remains a complex challenge. Cross-domain recommendation frameworks address this by seeking latent alignments between disparate item spaces. Zhu et al. proposed bridging these domains via a Cluster-Level Latent Factor Model, establishing that shared cluster-level factors can facilitate robust cross-domain translations [8]. These works underscore the necessity of achieving high-quality, domain-specific clusters before cross-domain recommendation can be effectively operationalized.

2.3 Explainable AI (XAI) in Recommendation

Explainability has transitioned from a supplementary feature to a core requirement. XAI methods require the underlying item representations to align with human reasoning. If a clustering algorithm’s structure is dominated by uninterpretable or imbalanced latent spaces, downstream explanation modules cannot generate accurate, faithful justifications for why an item was recommended.

3 Research Questions

Based on the challenges of high-dimensional feature representation in recommendation systems, this study investigates the following core research questions:

- **RQ1:** How does clustering on a unified global feature space compare to multi-view (isolated semantic subspace) clustering in terms of internal validation and structural stability?
- **RQ2:** To what extent do high-dimensional feature subsets (e.g., Tech & Comfort) mask the variance of lower-dimensional subsets (e.g., Ownership, Safety) in a global clustering pipeline?

4 Methodology

4.1 Dataset and Preprocessing

The empirical analysis was conducted on a proprietary dataset comprising $N = 251$ vehicle profiles (representing 249 unique models). Data was aggregated from public automotive databases—including Bharat NCAP and Global NCAP safety reports—and enriched with regional pricing and specification matrices. The dataset captures a diverse market covering 22 unique manufacturers and spanning multiple fuel types (predominantly Petrol), as detailed in Table 1. The

raw dataset encompasses a union of 115 features; however, exactly 114 features were utilized in the global clustering baseline (excluding one redundant feature during global concatenation). Prior to clustering, missing values were resolved using median imputation, followed by Z-score standardization per condition.

Table 1: Dataset Statistics Overview

Statistic	Value
Total Vehicles (Rows)	251 (249 unique models)
Unique Manufacturers	22
Ex-Showroom Price Range	5.49 to 296.00 Lakh
Mean Ex-Showroom Price	31.18 Lakh
SUV Body Styles	160
Sedan Body Styles	42
Hatchback Body Styles	30
MPV Body Styles	19

4.2 Feature Perspectives

To evaluate the impact of subset scoping, the global features were explicitly partitioned into four semantic perspectives representing diverse user purchasing intents (Note: the union across these perspectives totals 115 features):

- **Driving Performance (16 features)¹:** Variables related to physical dynamics.
- **Tech & Comfort (81 features)²:** High-dimensional variables detailing infotainment and convenience.
- **Ownership (7 features):** Variables representing the total cost of ownership.
- **Safety (11 features):** Standardized safety metrics and occupant protection scores.

¹The source specification and dimension tables yield 17 raw numeric fields for this perspective (engine displacement, cylinder count, power, torque, body dimensions, boot space, wheelbase, tyre geometry, rim diameter, ground clearance, kerb weight, fuel tank capacity, and turning radius). The engineered power-to-weight ratio, being a direct linear transform of two already-included features (power and kerb weight), was dropped as redundant prior to clustering, yielding the reported count of 16.

²The connectivity and convenience matrix contains 80 binary/ordinal feature indicators (excluding `model_name`) once the three pre-aggregated summary columns (Tech Score, Infotainment Score, Convenience Score) are removed to avoid double-counting information already captured by the underlying binary indicators. One of these three aggregate scores (Tech Score) was retained as a standalone engineered feature to preserve a holistic technology-adoption signal, bringing the perspective total to 81.

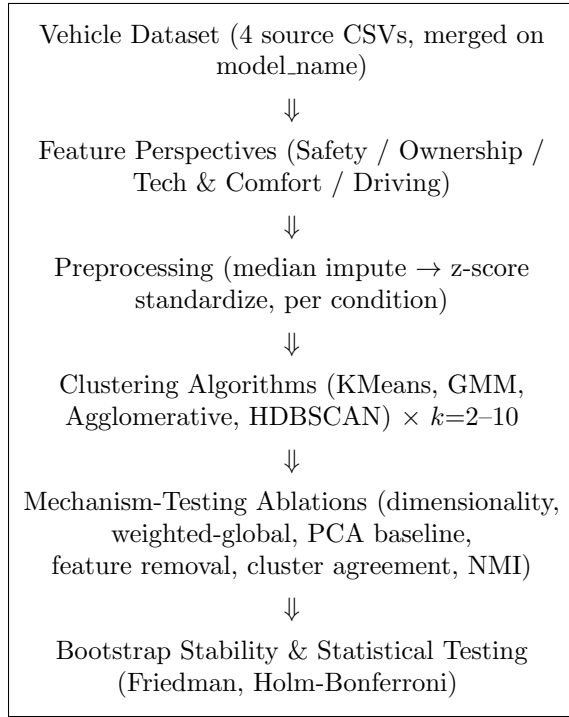


Figure 1: Experimental Pipeline Architecture

4.3 Clustering Algorithms and Hyperparameters

We evaluated four clustering paradigms to ensure our findings were not algorithm-specific (Table 2).

Table 2: Algorithm Configuration and Hyperparameters

Algorithm / Step	Hyperparameter	Value
KMeans	n_init	10
KMeans & GMM	random_state	42
GMM	covariance_type	full
Agglomerative	linkage	ward [6]
HDBSCAN	min_cluster_size	5 [9]
PCA (Baseline)	n_components	30
All	k_range	2–10
All	n_bootstrap	100

4.4 Evaluation Metrics

Models were evaluated using Silhouette Score [1], Davies-Bouldin Index [2], and Calinski-Harabasz Index [3]. The final k selection for each condition was determined by identifying the maximum mean silhouette score across the evaluated range. Stability was measured via the Adjusted Rand Index (ARI) and Adjusted Mutual Information (AMI) through 100 bootstrap iterations. We adopt Hennig’s (2007) framework [4], where $ARI \geq 0.80$ denotes a stable cluster and $ARI < 0.50$ indicates a dissolved cluster. Significance was assessed using Friedman tests blocked by k , followed by Holm-Bonferroni [5] corrected pairwise Wilcoxon signed-rank tests. Feature importance was quantified using Eta-squared (η^2), which measures the ANOVA-style proportion of variance in a given feature explained by the cluster assignments.

5 Experimental Results

5.1 Dimensionality and Subspace Masking

Pre-clustering variance analysis corroborates the foundation of the subspace masking hypothesis: under Z-score standardization, the Tech & Comfort perspective commands 70.43% of the global variance, while Ownership commands only 6.09%.

To test whether this dimensionality imbalance solely drives poor global clustering, we executed two baselines: a PCA-reduced global space (30 dimensions, 91.61% variance retained) and a variance-equalized `weighted_global` space. In the `weighted_global` operationalization, each perspective’s features were scaled by $1/\sqrt{D_{perspective}}$ before concatenation to guarantee that each semantic block

contributed equally to the total variance distance. The `weighted_global` condition improved the mean silhouette from 0.1695 (standard global) to 0.2118. The PCA condition yielded 0.1977. However, both still substantially underperformed the isolated Ownership perspective (0.4736) and Safety perspective (0.4602). This demonstrates that while feature imbalance contributes to masking, semantic entanglement intrinsically reduces cluster cohesion independent of feature-count balance.

Furthermore, a dimensionality ablation study revealed that real semantic clusters consistently outperformed randomly sampled feature subsets of identical dimensionality (e.g., Driving Performance real silhouette 0.3364 vs. random 0.3181 ± 0.0451).

5.2 Scope Comparison: Global vs. Perspective Clustering

Table 3 summarizes the optimal k and internal validation across scopes.

Table 3: Optimal Clustering Scope Summaries

Condition	Final k	Features	Mean Silhouette	HDBSCAN k
Global Full	2	114	0.1695	2
Weighted Global	5	115	0.2118	N/A
Global PCA	2	30	0.1977	N/A
Perspective: Safety	3	11	0.4602	17
Perspective: Ownership	7	7	0.4736	17
Perspective: Tech & Comf.	2	81	0.1995	4
Perspective: Driving Perf.	2	16	0.3252	14

The HDBSCAN baseline reveals a substantial reduction in observable density structure in the global space. While HDBSCAN identifies a $k = 2$ partition in the `global_full` condition, it partitions the data into 17 clusters within the localized Safety and Ownership subspaces. It should be acknowledged, however, that the `min_cluster_size=5` hyperparameter on a modest sample of $N = 251$ mechanically favors the discovery of smaller groupings; thus, while $k = 17$ highlights localized density, it is partially a hyperparameter artifact.

5.3 Cluster Independence and Agreement

To quantify representational overlap, we calculated the Adjusted Rand Index (ARI) mapping perspective clusters back to the global clusters. The Tech & Comfort clusters exhibited near-perfect redundancy with the global clusters (ARI = 0.8911). In contrast, Ownership (ARI = 0.0944) and Safety (ARI = 0.1167) were weakly correlated with the global structure. Removing the Tech & Comfort features from the global space substantially increased Safety’s cluster agreement (ARI increased to 0.2436), confirming that high-dimensional perspectives mathematically suppress localized signals.

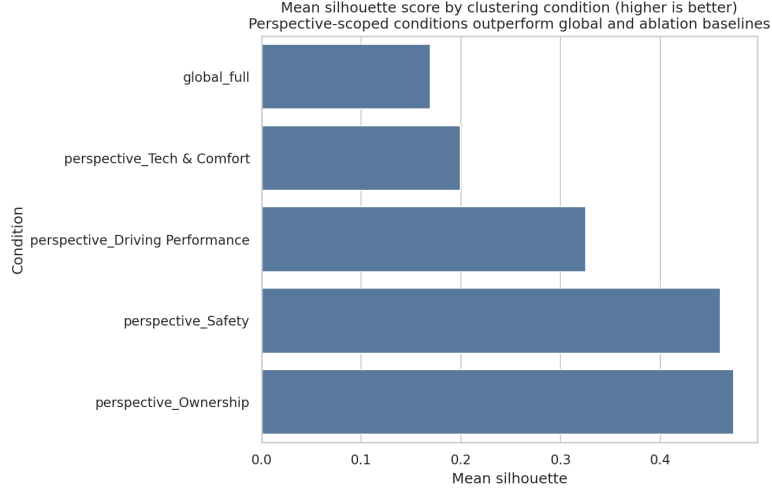


Figure 2: Comparison of Mean Silhouette Scores. The perspective-specific clusters (Safety, Ownership) outperform the standard and ablated global spaces, highlighting the structural advantages of localized semantic framing.

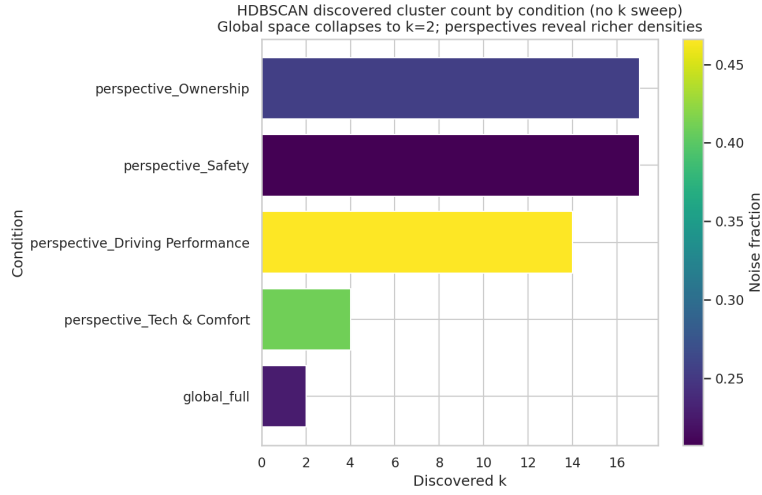


Figure 3: Clusters Discovered by HDBSCAN. The global feature space yields $k = 2$, whereas localized perspectives exhibit broader density variations ($k = 14$ to 17).

Cross-perspective Normalized Mutual Information (NMI) further corroborates this functional independence. Safety versus Ownership yielded an NMI of just 0.1678, suggesting each perspective encodes a distinct facet of the vehicle.

5.4 Stability, Balance, and Statistical Significance

Bootstrap stability indicates higher robustness for localized perspectives. Using Hennig’s thresholds, KMeans on the Safety perspective ($k = 3$) achieved an ARI of 0.9880 (highly stable). Conversely, Agglomerative clustering on the global space yielded an ARI of 0.4349, placing it in the unstable tier.

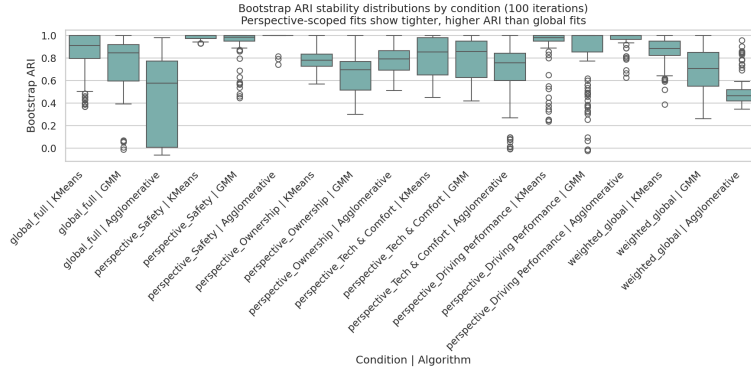


Figure 4: Bootstrap Stability Distributions (100 iterations). Models fit on semantic perspectives achieve higher Adjusted Rand Index (ARI) scores than the global condition.

Cluster balance, measured via normalized entropy, remained exceptionally high (≈ 0.99) for global $k = 2$ models, but demonstrated greater variance (e.g., Gini = 0.5020 for Ownership GMM $k = 7$) in the higher- k perspective models, reflecting unevenly distributed vehicle archetypes.

5.5 Feature Interpretability

Analysis of feature importance metrics reveals a substantial imbalance in subspace influence during global clustering.

The top distinguishing features for the `global_full` clusters are predominantly derived from the Tech & Comfort domain (e.g., Auto Hold $\eta^2 = 0.7859$). When clustered independently, the isolated perspectives expose their respective domain-specific discriminators (e.g., Ownership is partitioned by unit fuel price, $\eta^2 = 0.9839$).

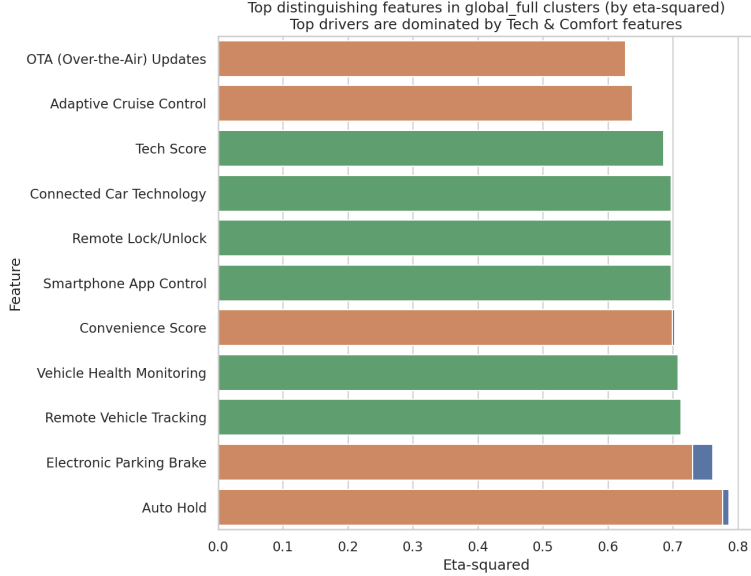


Figure 5: Global Feature Importance (η^2). The top distinguishing features driving the `global_full` partitions are derived from the Tech & Comfort domain.

6 Discussion

The empirical results provide strong evidence supporting the subspace masking hypothesis. They expose how different mathematical clustering assumptions behave when confronted with heterogeneous automotive data. By analyzing algorithm-specific outputs, we can draw distinct domain conclusions regarding representation learning in vehicle markets.

6.1 Euclidean Bias and the “Technology-Rich vs. Technology-Sparse” Split (KMeans)

In the `global_full` condition, KMeans systematically favored a $k = 2$ partition, driven largely by the Tech & Comfort perspective (e.g., Auto Hold $\eta^2 = 0.785$, Electronic Parking Brake $\eta^2 = 0.761$). Because KMeans relies on unweighted Euclidean distance, the 81 technology features heavily influence the geometry of the space. In the context of the automotive domain, this means KMeans is primarily slicing the market into a “technology-rich” cluster and a “technology-sparse” cluster. While this is a mathematically valid partition, it may lack deeper semantic utility. It largely overlooks whether a vehicle is a high-performance sports car or a heavy commercial utility vehicle, so long as both lack similar advanced infotainment features.

6.2 Distributional Assumptions and Market Realities (Agglomerative & GMM)

Agglomerative clustering utilizing Ward’s linkage achieved the lowest global stability ($ARI = 0.434$). Ward’s linkage assumes that clusters are roughly spherical and exhibit equal variance. However, the automotive market is not uniformly distributed; economy vehicles often share highly similar feature profiles, while specialized vehicles may be sparsely distributed. When the 81 Tech & Comfort features apply unequal variance pressure across the hierarchical tree, the assumption of approximately spherical clusters appears inconsistent with the observed data distribution, resulting in unstable structural recovery across bootstrap iterations.

Similarly, Gaussian Mixture Models (GMM) struggled in the global space. Fitting full covariance matrices for 114 dimensions on only 251 vehicles triggers the curse of dimensionality. The model becomes starved for data, increasing the risk of fitting spurious covariance structure to the technology features rather than identifying true latent automotive archetypes.

6.3 Density Differences in Automotive Segments (HDBSCAN)

The HDBSCAN baseline offers additional evidence consistent with the subspace masking hypothesis. In lower-dimensional subspaces like Ownership (7 features) and Safety (11 features), HDBSCAN discovered highly specific topological densities ($k = 17$). In the automotive domain, these localized clusters may correspond to true market segments (e.g., high-safety family SUVs, low-cost commuter hatchbacks). However, when mapped into the 114-dimensional global space, these localized density structures are no longer resolved. The distance between points expands uniformly, causing the market segments to merge into a single broad cluster with isolated noise ($k = 2$). These results suggest that concatenating heterogeneous feature domains obscures local structure that is preserved in perspective-specific representations.

6.4 Implications for Explainable Automotive Recommendation

Explainable recommender systems must map items to specific user intents. If a user seeks a vehicle recommendation anchored in safety or cost-to-own priorities, a monolithic global clustering pipeline may struggle to generate contextually accurate results. These findings indicate that modern automotive recommender systems may benefit from transitioning away from monolithic feature representations. Instead, deploying an ensemble of perspective-specific clusters ensures that recommendations remain grounded in the semantic domain the user cares about, potentially allowing downstream Large Language Models (LLMs) to generate more faithful, intent-aligned explanations.

7 Threats to Validity

Construct Validity: While we utilized established internal validation metrics, distance-based clustering assumes uniform feature scaling. Furthermore, statistical significance testing relied on a Friedman chi-square approximation, which may suffer from reduced power given our low block count ($n_{blocks} \leq 15$).

8 Limitations

While this empirical benchmark rigorously isolates the structural mechanics of subspace masking, several limitations must be acknowledged:

- **Single Dataset Constraint:** A primary scientific limitation of this study is its reliance on a single dataset ($N = 251$). While the dataset represents a cross-section of the market, the conclusions regarding variance masking and topological constraints are inherently bounded by the distributions of this specific corpus. Expanding this evaluation across multiple distinct automotive datasets is necessary to generalize these architectural findings fully.
- **Sample Size to Dimensionality Ratio:** Mapping a restricted sample size against a high-dimensional global feature space ($D = 114$) inherently challenges density-based and probabilistic models. While techniques like PCA and variance-weighting were applied to mitigate this, fitting full-covariance Gaussian Mixture Models on such sparse matrices risks ill-conditioning.
- **Absence of Categorical and Brand Variables:** The clustering models currently operate exclusively on median-imputed, Z-score standardized numerical and ordinal features. However, real-world automotive purchasing decisions are often constrained by categorical topologies (e.g., body style: SUV versus Sedan) and brand equity. Although documented in our metadata, the exclusion of these non-numerical dimensions from the distance matrices limits the immediate mapping of these clusters to complete consumer personas.
- **Proxy Metrics for Downstream Utility:** We rely on internal validation metrics and structural stability to evaluate cluster quality. However, internal cohesion is only a proxy for recommendation utility. This study does not perform human-in-the-loop evaluations to confirm that perspective-specific clusters yield higher user satisfaction or more faithful LLM-generated explanations in a production environment.

9 Conclusion

Automotive recommender systems frequently rely on monolithic feature representations, concatenating complex and semantically distinct attributes—such

as safety ratings, technology, and ownership costs—into a single mathematical space. This empirical study establishes that subspace masking is an architectural phenomenon that emerges from these heterogeneous feature representations, and one that can be empirically isolated through controlled ablation. By benchmarking a dataset of 251 vehicles, we provide strong evidence that high-dimensional feature subsets can overpower and obscure the localized topological structures of lower-dimensional domains.

Answering RQ1, our results demonstrate that perspective-specific clustering offers a more robust alternative to global clustering pipelines. When vehicles are clustered within isolated semantic bounds, algorithmic stability increases substantially—frequently exceeding ARI thresholds of 0.90—and internal cluster cohesion improves. Answering RQ2, our mechanism-testing ablations—including PCA-reduced baselines, weighted-global representations, and feature-removal studies—confirm that this behavior is not merely a byproduct of reduced dimensionality, but a consequence of reducing cross-domain feature interference. Algorithms like HDBSCAN indicate that natural densities exist within individual domains, but these densities are obscured when represented in a concatenated Euclidean geometry.

The architectural implications of these findings are substantial. If recommender systems are expected to generate transparent, user-aligned suggestions, the underlying data structures must reflect the specific intent of the user. A system may struggle to provide a faithful safety-oriented recommendation if its latent clusters are primarily defined by infotainment features.

While this study is constrained by its evaluation of a single dataset, it provides an empirical blueprint for representation learning in explainable AI (XAI). Moving forward, transitioning away from global representations in favor of semantic, multi-view clustering architectures presents a promising pathway for the development of context-aware automotive recommendation engines. Subsequent research should prioritize evaluating how these structural improvements actively translate into the quality and trustworthiness of downstream, natural-language LLM explanations.

Code Availability

To support reproducibility, the complete implementation accompanying this study is publicly available at:

https://github.com/mazcter/Tabless_wheels

The repository contains the complete experimental pipeline, including data preprocessing, perspective-specific feature construction, clustering implementations, benchmarking scripts, bootstrap stability analysis, statistical testing, and the utilities required to reproduce the figures and tables presented in this paper.

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